Velocity field in a Newtonian viscous half-space driven by a prescribed surface *horizontal* velocity

Problem

A constant-viscosity (μ), incompressible Stokes half-space occupies $z \geq 0$, with z pointing *downward* into the medium and the surface at $z = 0$. A distribution of *horizontal* (tangential) velocity

$$v_x(x, 0) = V(x)$$

is prescribed at the surface. The companion condition is a vanishing *normal* response, $v_z(x, 0) = 0$ (impermeable surface); for the decaying half-space modes this coincides with the free-surface condition $\sigma_{zz}(x, 0) = 0$, so the two give the same interior field. Velocities vanish as $z \to \infty$. The governing equations are

$$\nabla \cdot \boldsymbol{v} = 0, \qquad \mu \, \nabla^2 \boldsymbol{v} = \nabla p, \qquad \nabla^4 \psi = 0,$$

with stream function $v_x = \partial_z \psi$, $v_z = -\partial_x \psi$.

Convolution formulas (real space)

The interior velocity field is obtained by convolving the surface datum V with two Poisson-type kernels:

$$\boxed{v_x(x, z) = \int_{-\infty}^{\infty} \frac{2\,(x - x')^2\, z}{\pi \left[(x - x')^2 + z^2\right]^2} \, V(x') \, \mathrm{d}x',} \tag{1}$$

$$\boxed{v_z(x, z) = \int_{-\infty}^{\infty} \frac{2\,(x - x')\, z^2}{\pi \left[(x - x')^2 + z^2\right]^2} \, V(x') \, \mathrm{d}x'.} \tag{2}$$

Equivalently, $v_x = K_x * V$ and $v_z = K_z * V$ with kernels

$$K_x(x, z) = \frac{2\, x^2\, z}{\pi\, (x^2 + z^2)^2}, \qquad K_z(x, z) = \frac{2\, x\, z^2}{\pi\, (x^2 + z^2)^2}.$$

Properties. K_x is even in x, K_z is odd in x, and

$$\int_{-\infty}^{\infty} K_x(x, z) \, \mathrm{d}x = 1, \qquad \int_{-\infty}^{\infty} K_z(x, z) \, \mathrm{d}x = 0.$$

A spatially uniform V therefore produces a uniform horizontal flow and no vertical flow. A localized V (a patch of tangential drag) drives convergence/downwelling ahead of the patch and divergence/upwelling behind it. At the surface $v_z(x, 0) = 0$: the imposed drag produces no normal surface motion.

Reciprocity. The cross-kernel K_z here is identical to the horizontal-response kernel obtained when a *vertical* surface velocity is imposed. The vertical response to horizontal forcing equals the horizontal response to vertical forcing, as required by the symmetry of the Stokes operator.

Spectral form (transfer functions)

With the Fourier convention $\hat{f}(k) = \int f(x)\, e^{-ikx}\, \mathrm{d}x$ and $q \equiv |k|$, the convolutions are diagonal multiplications:

$$\hat{v_x}(k,z) = (1 - qz)\, e^{-qz}\, \hat{V}(k), \qquad \hat{v_z}(k,z) = -\, ik\, z\, e^{-qz}\, \hat{V}(k).$$

The associated pressure and the surface *shear* traction are

$$\hat{p}(k,z) = -\, 2\mu\, ik\, e^{-qz}\, \hat{V}(k), \qquad \hat{\sigma}_{xz}(k,0) = -\, 2\mu\, |k|\, \hat{V}(k),$$

while $\hat{\sigma}_{zz}(k,0) = 0$. In real space the driving shear traction is $\sigma_{xz}(x,0) = -2\mu\, \Lambda\, V(x)$ with $\Lambda = \sqrt{-\partial_x^2}$, the exact tangential counterpart of the $-2\mu|k|$ normal admittance of the vertical-drive problem. Numerically: FFT $V(x)$, multiply by the two transfer functions above, and inverse-FFT to recover v_x and v_z.

Sanity check: a spreading surface gives corner flow

Take the piecewise-constant datum

$$V(x) = V_0\, \mathrm{sgn}(x),$$

i.e. the surface is dragged *outward* at speed V_0 on either side of $x = 0$ — the kinematics of two diverging plates. Evaluating the convolution integrals (equivalently, the spectral solution with $\widehat{\mathrm{sgn}}(k) = 2/ik$) gives the closed form

$$\psi(x,z) = \frac{2V_0}{\pi}\, z\, \arctan\frac{x}{z},$$

$$v_x(x,z) = \frac{2V_0}{\pi}\left(\arctan\frac{x}{z} - \frac{xz}{x^2+z^2}\right), \qquad v_z(x,z) = -\,\frac{2V_0}{\pi}\,\frac{z^2}{x^2+z^2}.$$

As $z \to 0^+$, $\arctan(x/z) \to \frac{\pi}{2}\,\mathrm{sgn}(x)$ and the second term vanishes, so $v_x \to V_0\,\mathrm{sgn}(x)$ and $v_z \to 0$: the boundary datum is recovered.

Self-similarity. In polar coordinates $x = r\cos\phi,\ z = r\sin\phi$ with $\phi \in (0,\pi)$ measured from the surface, $\arctan(x/z) = \frac{\pi}{2} - \phi$, so

$$\psi = \frac{2V_0}{\pi}\, r\, \sin\phi\, \left(\tfrac{\pi}{2} - \phi\right) = r\Big(V_0 \sin\phi - \tfrac{2V_0}{\pi}\,\phi\sin\phi\Big).$$

The stream function is homogeneous of degree one in r, and ψ/r is the standard combination of the degree-one biharmonic modes $\sin\phi$ and $\phi\sin\phi$: both velocity components depend on ϕ alone. This is precisely the classical two-dimensional Stokes *corner flow*.

Axial upwelling. On the axis $x = 0$ the flow is purely vertical, with the depth-independent upwelling speed

$$\big|v_z(0,z)\big| = \frac{2}{\pi}\, V_0,$$

This means we have an initial condition for velocity in z?

the textbook passive-upwelling rate beneath a spreading centre whose half-rate is V_0.

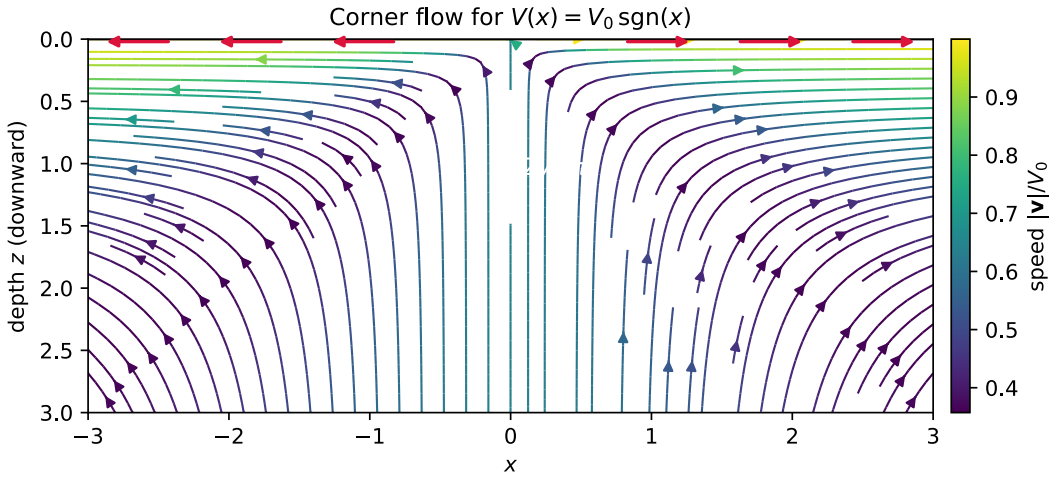


Figure 1: Streamlines of the field driven by $V(x) = V_0\,\mathrm{sgn}(x)$ (colour: speed). Red arrows: the diverging surface velocity. The passive return flow wells up along the axis at the depth-independent rate $2V_0/\pi$ (white arrow) and turns to feed the outgoing surface layer — the self-similar corner flow of a spreading centre.

Three-dimensional case

For a surface datum $\boldsymbol{V}(x, y) = \big(V_x, V_y, 0\big)$ on the plane $z = 0$, with $\boldsymbol{k} = (k_x, k_y)$ and $q = |\boldsymbol{k}|$, the vertical structure functions $(1 - qz)\,e^{-qz}$ and $-iz\,e^{-qz}$ carry over, but the response now has a tensor character: it must be resolved into the components of $\boldsymbol{V}$ parallel and perpendicular to $\boldsymbol{k}$, only the parallel part feeding the vertical velocity and pressure through continuity. The spectral multiply–then–inverse-FFT route remains the practical method; a single closed-form real-space kernel of the 2D type above no longer suffices.